

Overview

A new test file `packages/graph-core/src/__tests__/schema.test.ts` (added lines R1-730) exercises the graph-schema utilities: `validateGraph`, `sanitizeGraph`, `autoFixGraph`, and the alias maps `NODE_TYPE_ALIASES` / `EDGE_TYPE_ALIASES`. The file imports `vitest` helpers, the core functions, the alias constants, and the `KnowledgeGraph` type.

Key changes

- **Imports:** `vitest` helpers and the schema utilities are added (R1-R6).
- **Fixture:** `validGraph` (lines 12-59) provides a minimal, well-formed graph used throughout the tests.
- **Validation tests:** cover success, missing required fields, invalid node/edge types, dangling references, alias normalization (e.g., `"func"` → `"function"`), and enforcement that alias values are not alias keys (lines 61-268).
- **Sanitization tests:** verify null-to-undefined conversion, lowercasing of enum-like strings, and handling of null `tour` / `layers` (lines 280-349).
- **Auto-fix tests:** check defaulting of missing fields, alias mapping, type coercion, and out-of-range clamping (lines 351-500).
- **Extended type tests:** confirm support for new node types (`config`, `document`, ...) and edge types (`deploys`, `serves`, ...), including legacy alias resolution (lines 502-730).
- **Permissive validation:** tests that nodes/edges with missing IDs or references are dropped with appropriate issues, and fatal errors are raised when no valid nodes remain (lines 504-590).

Impact

- **Coverage:** The test suite adds numerous assertions that exercise edge cases and alias handling.
- **Maintainability:** Changes to alias maps or validation logic will be caught early.
- **Compatibility:** Requires `vitest` as a dev dependency; no runtime changes.

Risks & follow-ups

- **Regression:** Altering alias constants or validation rules may cause failures; review these constants.
- **Test brittleness:** The shared `validGraph` fixture must be deep-cloned (`structuredClone`) to avoid accidental mutation.
- **Coverage gaps:** Tests focus on positive paths; consider adding negatives for malformed input types.

- **Performance:** The suite contains many assertions; monitor runtime to keep it acceptable.